

IOAI Philippines 2026: National Semi-Finals (Theory Assessment)

Welcome, Semi-Finalist!

You have proven your ability to build models in the Practical Round. Now, we must verify your understanding of the mathematical and theoretical foundations of Artificial Intelligence, as required by the **IOAI Official Syllabus**.

This assessment covers the "**Theory**" and "**Both**" categories of the syllabus, including Classical ML, Deep Learning mechanics, Computer Vision, and NLP.

Important Email Instruction:

Please ensure you submit this form using **ONLY** your **Registered Email Address** or the email address associated with your **Kaggle account**. Submissions from unrecognized emails may not be matched to your leaderboard score and could result in disqualification.

Exam Rules:

- **Format:** 20 Multiple Choice Questions.
- **Window:** Open from **Feb 14 (6:00 PM)** to **Feb 16 (10:59 AM)**.
- **Time Limit:** None. You may take as much time as needed within the submission window.
- **Protocol: Open-Book.** Given the remote nature of this round, you may consult standard documentation and textbooks. However, collaboration with other students or asking AI to directly solve the questions is strictly prohibited.
- **Scoring:** This exam constitutes **30%** of your Round 2 Grade.

The "Safety Floor" Rule:

To qualify for the **Top 20 (National Finals)**, you must achieve a minimum score of **10/20 (50%)** on this exam. High practical scores on Kaggle will **not** save you if you cannot demonstrate fundamental theoretical competence.

Good luck. The next step is the National Finals.

* Indicates required question

1. Email *
-

2. **Academic Integrity:** By submitting this form, you affirm that the answers provided are your own work and that you have not collaborated with others to answer these questions. *

Check all that apply.

☐ I affirm.

Part 1: Foundational Skills & Classical ML

3. **Data Processing (Sensor Noise):** You are given a weather dataset where the `relative_humidity_2m` column contains several entries of 999%. Based on standard data processing practices for sequence modeling, what is the most appropriate first step? *

Mark only one oval.

- ☐ A) Impute the values with the mean of the entire column.
- ☐ B) Clip the values to a maximum of 100% or replace with NaN for imputation.
- ☐ C) Leave the values as-is to allow the model to learn the sensor error.
- ☐ D) Drop the entire humidity column to avoid bias.

4. **Feature Engineering (Cyclic Time):** When modeling seasonal rainfall, why is it mathematically superior to represent the "Day of the Year" using `sin_year` and `cos_year` features rather than a single integer (1–365)? *

Mark only one oval.

- ☐ A) It reduces the memory footprint of the dataset.
- ☐ B) It prevents the model from treating December 31 and January 1 as distant values.
- ☐ C) It allows the model to ignore leap years.
- ☐ D) It is a required step for Gradient Boosting models.

5. **Feature Engineering (Proxies):** In the absence of a future rainfall label, a student uses soil_moisture_0_to_7cm as a primary feature. Why is this considered a "strong proxy" for precipitation in a weather model? *

Mark only one oval.

- ☐ A) Soil moisture is a leading indicator that causes rain.
- ☐ B) Soil moisture reacts instantly to rainfall, capturing the "physical footprint" of a rain event.
- ☐ C) It is the only variable in the syllabus that uses metric units.
- ☐ D) It is less noisy than temperature data.

6. **Supervised Learning (Regularization):** Which statement correctly describes the difference between L1 (Lasso) and L2 (Ridge) regularization? *

Mark only one oval.

- ☐ A) L1 regularization can shrink coefficients to exactly zero, performing feature selection.
- ☐ B) L2 regularization is only used for classification, not regression.
- ☐ C) L1 regularization is generally better at handling high-variance data.
- ☐ D) L2 regularization always results in a simpler, more interpretable model.

7. **Unsupervised Learning (K-Means):** When using K-Means clustering to identify regional climate zones in the Philippines, how is the "Elbow Method" typically used? *

Mark only one oval.

- ☐ A) To determine the optimal learning rate for the centroids.
- ☐ B) To find the number of clusters (K) where the marginal gain in variance explained begins to level off.
- ☐ C) To visualize the clusters in a 2D space using t-SNE.
- ☐ D) To remove outliers from the training set.

Part 2: Neural Networks & Deep Learning

8. **Neural Networks (Backpropagation):** The fundamental mathematical principle that allows gradients to be calculated from the output layer back to the input layer is: *

Mark only one oval.

- ☐ A) The Law of Large Numbers.
- ☐ B) The Chain Rule of Calculus.
- ☐ C) The Central Limit Theorem.
- ☐ D) Bayesian Inference.

9. **Activation Functions:** In a Deep Multi-Layer Perceptron (MLP), why is the **ReLU** activation function generally preferred over **Sigmoid**? *

Mark only one oval.

- ☐ A) ReLU is computationally more expensive but more accurate.
- ☐ B) ReLU helps mitigate the "vanishing gradient" problem in deep layers.
- ☐ C) ReLU ensures that the output is always between 0 and 1.
- ☐ D) Sigmoid can only be used in the final output layer.

10. **Regularization (Dropout):** During the training of a neural network, what is the primary mechanism by which "Dropout" prevents overfitting? *

Mark only one oval.

- ☐ A) It reduces the learning rate automatically when validation loss stalls.
- ☐ B) It randomly deactivates neurons, forcing the network to learn redundant representations.
- ☐ C) It adds a penalty to the loss function based on the magnitude of the weights.
- ☐ D) It increases the number of training samples through augmentation.

11. **Attention Mechanisms:** The "Scaled Dot-Product Attention" used in Transformers relies on which three primary components? *

Mark only one oval.

- ☐ A) Inputs, Hidden States, and Outputs.
- ☐ B) Kernels, Strides, and Padding.
- ☐ C) Queries, Keys, and Values.
- ☐ D) Encoders, Decoders, and Embeddings.

12. **Model Fine-Tuning:** What is the primary advantage of **Parameter-Efficient Fine-Tuning (PEFT)** compared to full fine-tuning of a Large Language Model? *

Mark only one oval.

- ☐ A) It allows the model to exceed its original context window.
- ☐ B) It requires updating only a small fraction of the model's weights, saving significant compute.
- ☐ C) It eliminates the need for any labeled training data.
- ☐ D) It is the only way to adapt a model for Computer Vision tasks.

Part 3: Computer Vision & NLP

13. **CV (CNN Layers):** What is the primary purpose of a **Pooling Layer** (e.g., Max Pooling) in a Convolutional Neural Network? *

Mark only one oval.

- ☐ A) To increase the number of channels in the feature map.
- ☐ B) To reduce the spatial dimensions of the feature map, reducing parameters and computation.
- ☐ C) To apply a non-linear activation to the pixels.
- ☐ D) To detect specific edges and textures in the image.

14. **CV (Object Detection):** Which of the following architectures is specifically designed for single-pass, real-time object detection? *

Mark only one oval.

- ☐ A) ResNet
- ☐ B) BERT
- ☐ C) YOLO (You Only Look Once)
- ☐ D) U-Net

15. **CV (Segmentation):** In a weather monitoring system, which model architecture would be most appropriate for "masking" or outlining the exact area of a satellite image covered by a tropical depression? *

Mark only one oval.

- ☐ A) VGG-16
- ☐ B) U-Net
- ☐ C) K-Nearest Neighbors
- ☐ D) Logistic Regression

16. **NLP (Transformers):** In the context of NLP, what does the "Self-Attention" mechanism allow a model to do? *

Mark only one oval.

- ☐ A) Focus on each word in a sequence relative to every other word to capture context.
- ☐ B) Memorize the entire training vocabulary.
- ☐ C) Translate text without using a decoder.
- ☐ D) Automatically correct grammatical errors in the input.

17. **NLP (BERT):** Which training objective is a core part of the original BERT (Bidirectional Encoder Representations from Transformers) model? *

Mark only one oval.

- ☐ A) Next Sentence Prediction (NSP).
- ☐ B) Image-to-Text Generation.
- ☐ C) K-Means Clustering of word vectors.
- ☐ D) Reinforcement Learning from Human Feedback.

Part 4: Data Science & Evaluation

18. **Evaluation Metrics:** If your model is predicting rainfall in millimeters (mm), why might you choose **MAE (Mean Absolute Error)** over **RMSE (Root Mean Squared Error)** as your primary metric? *

Mark only one oval.

- ☐ A) MAE is easier to calculate by hand.
- ☐ B) MAE is less sensitive to extreme outliers (heavy storm events) than RMSE.
- ☐ C) MAE always produces a lower numerical value than RMSE.
- ☐ D) RMSE can only be used for classification problems.

19. **Model Evaluation:** A student reports a Training MAE of **0.02** but a Validation MAE of **0.25**. What is the most likely diagnosis of this model? *

Mark only one oval.

- ☐ A) Underfitting (the model is too simple).
- ☐ B) Overfitting (the model has memorized the training data).
- ☐ C) The learning rate is too low.
- ☐ D) The dataset is too large.

20. **Data Processing (Scaling):** When features have very different scales (e.g., Pressure in hPa vs. Temperature in Celsius), why is **RobustScaler** often preferred over **StandardScaler**? *

Mark only one oval.

- ☐ A) It scales data to a range of exactly 0 to 1.
- ☐ B) It uses the median and interquartile range, making it less influenced by extreme sensor outliers.
- ☐ C) It is the only scaler compatible with XGBoost.
- ☐ D) It automatically handles missing NaN values.

21. **Deep Learning (Optimizers):** The **Adam** optimizer is widely used because * it combines the benefits of which two other concepts?

Mark only one oval.

- ☐ A) L1 and L2 Regularization.
- ☐ B) Momentum and Adaptive Learning Rates (RMSProp).
- ☐ C) Dropout and Early Stopping.
- ☐ D) Bagging and Boosting.

22. **Foundational Skills (Programming):** In Python (Pandas), which command * is used to find the number of missing values in each column of a weather dataset?

Mark only one oval.

- ☐ A) df.describe()
- ☐ B) df.isnull().sum()
- ☐ C) df.groupby('city_name').count()
- ☐ D) df.dropna()

This content is neither created nor endorsed by Google.

Google Forms

